



Erasmus Mundus WAVES

Acoustic Imaging Lab

Time-Reversal Acoustics

Submitted By: Group F

Harsh Christopher Singh

Janice Novi Prasetyo

Subarna Paudel

Work developed in the scope of the course

Acoustic Imaging

2025

Introduction

Time Reversal (TR) acoustics is a signal processing technique that focuses wave energy to a selected point in space and time. It started in the 1960s as a technique for signal transmission in the ocean between ships. It has been used for underwater acoustic communications, biomedical lithotripsy using focused ultrasound, earthquake localization and characterization, non-destructive evaluation for detecting and imaging cracks and defects in structures and for focusing sound on rooms.

The reversibility of the acoustic wave equation serves as the foundation for the fundamental idea of time reversal. Waves that radiate from a source and are captured by a variety of transducers may retrace their travels and constructively interfere at the original source position if the recorded signals are reversed in the time domain (played backwards) and re-emitted into the medium. As a result, acoustic energy is concentrated both spatially and temporally.

Objectives:

1. To understand the theoretical foundations of time reversal acoustics.
2. To perform the time of reversal wave reconstruction at the actual source location for a MEMS microphone array for two acquisition time:
 - a) short acquisition time: capturing direct wave contribution
 - b) long acquisition time: capturing reflected waves
3. To perform spatial focalization by backpropagating the time reversed signal at a location near the actual source location.

Theory

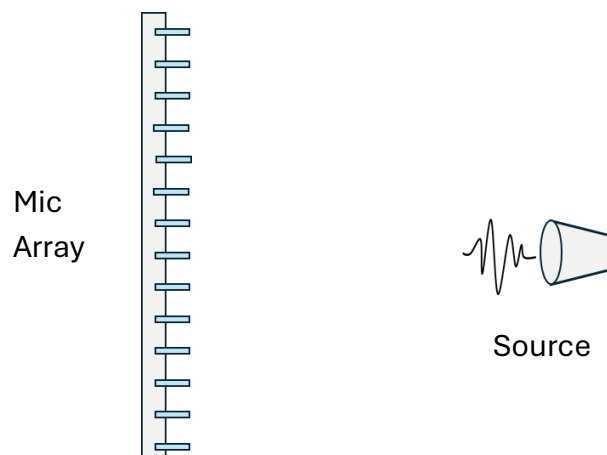
The Time-Reversal Mirror (TRM) is made up of a series of transducers that record, time-reverse, and retransmit acoustic signals.

- **Closed TRM:** A closed chamber completely covered with transducers to capture the whole wavefield (pressure and normal velocity) would be ideal for accurately reconstructing the source.
- **Open TRM:** In reality, a small aperture is frequently implied, or an open mirror. Placing the TRM in a complex, reverberant medium might paradoxically increase resolution, even if this restricts the resolution in open space (diffraction limit). The room's numerous reflections serve as virtual sources, therefore expanding the mirror's aperture.

The MEMS microphones utilized for recording in this experiment are unable to re-emit sound. As a result, the experiment depends on the **Principle of Superposition** and the **Reciprocity Theorem**.

Reciprocity principle: The field propagating from a source at $\mathbf{r}_s$ to a receiver at $\mathbf{r}_r$ is identical to the field transmitted from a source at $\mathbf{r}_r$ measured at $\mathbf{r}_s$.

In this experiment, the source is either transferred to the receiver positions (virtually) rather than physically re-emitting from the array, or, as in this experiment, the source remained at $\mathbf{r}_{s3}$ and the route is considered reciprocal. The backpropagation is captured by transmission from the source, recording at the recipient, and computationally adding the contributions.



In complex media (Lossy medium): Time reversal may not focus correctly on a lossy medium due to attenuation breaking the time-reversal invariance. However, the effective aperture determines the resolution in a reverberant (reflection) environment with little loss. Because the reverberating multipath signals include high spatial frequency of information that would be lost in a free-field open mirror arrangement, TR exhibits the super-resolution phenomena.

Calculations

The following preliminary calculations were done to check the validity of the setup.

For far-field calculation:

Frequency: [3,5] kHz

Central frequency (f_c): 4213 Hz

Source (D_s): 9 cm

Distance = $(2D_s^2) / \lambda = 0.2$ m, thus source and array distance should be greater than 20 cm.

Spatial Sampling:

Pitch length: 2.5 cm

Maximum possible frequency: $f_{\max} = c/2 \cdot d = 343/2/0.025 = 6860$ Hz.

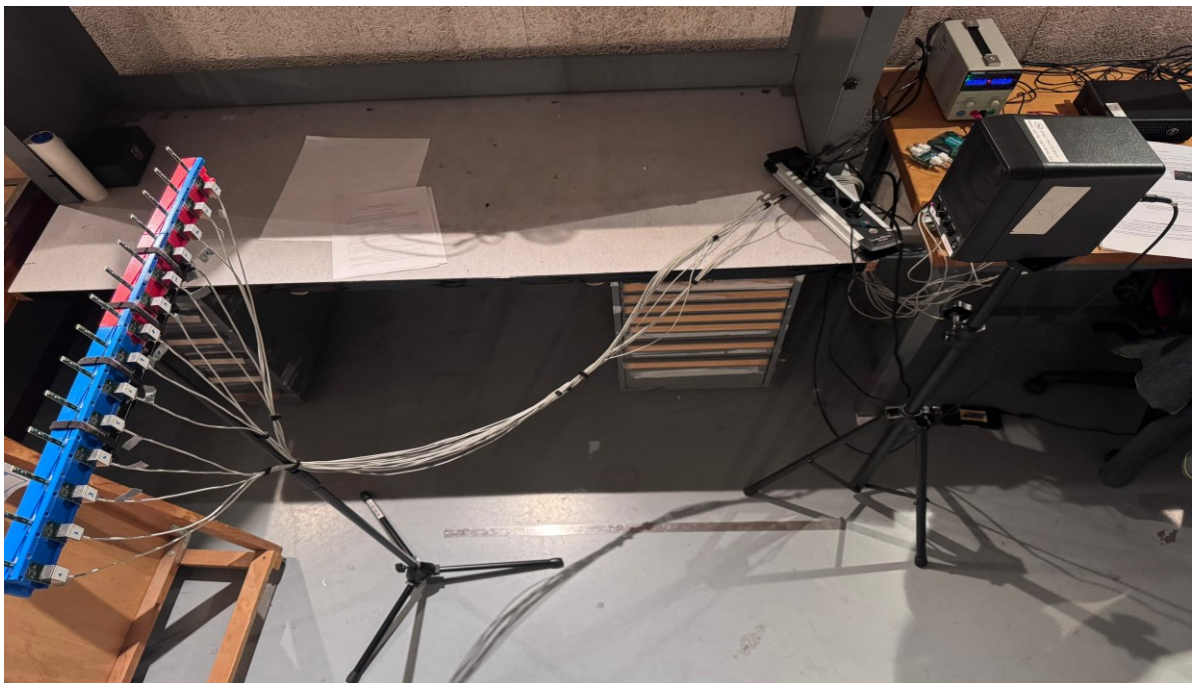
As the source has Ricker signal of central frequency 4213 Hz, it satisfies the sampling criteria.

Methodology

This experiment was carried out in a reflective room. This facilitates the observation and analysis of the effect of environment complexity (mainly reverberation) on the time reversal wave reconstruction at the actual source location.

Tools Used:

- Source: Loudspeaker transmitting Gaussian pulse, $f = 4213$ Hz for 4 ms.
- Receiver: Linear array of MEMS microphones (50 Hz – 20 kHz).
- Data Acquisition system: AD Discovery box (Analog) to generate the source signal and DD Discovery box (Digital) for capturing the MEMS Mic data via I2S protocol.
- Software: Digilent WaveForms (wavegen) and MATLAB for signal processing.



Procedures:

1. Forward propagation of Signal:
 - Source position is first kept at $d_1 < 1$ m and second $d_2 > 1$ m.
 - The Gaussian pulse is loaded and sent with a 250 Hz trigger for 4 ms.

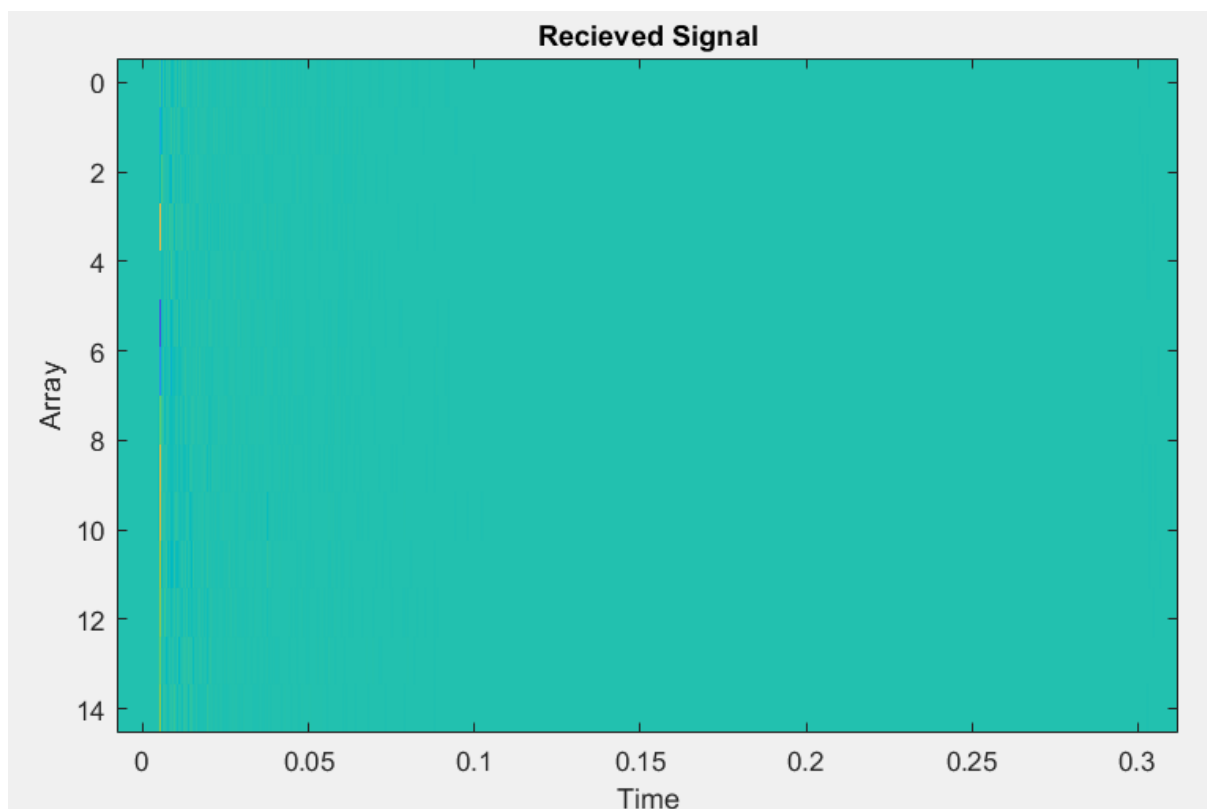
- Response was recorded by MEMS mic. It measures the impulse response (Green's function) between the source and each receiver.
2. Time Reversal:
- Recorded signals are time-reversed and saved in .mat files.

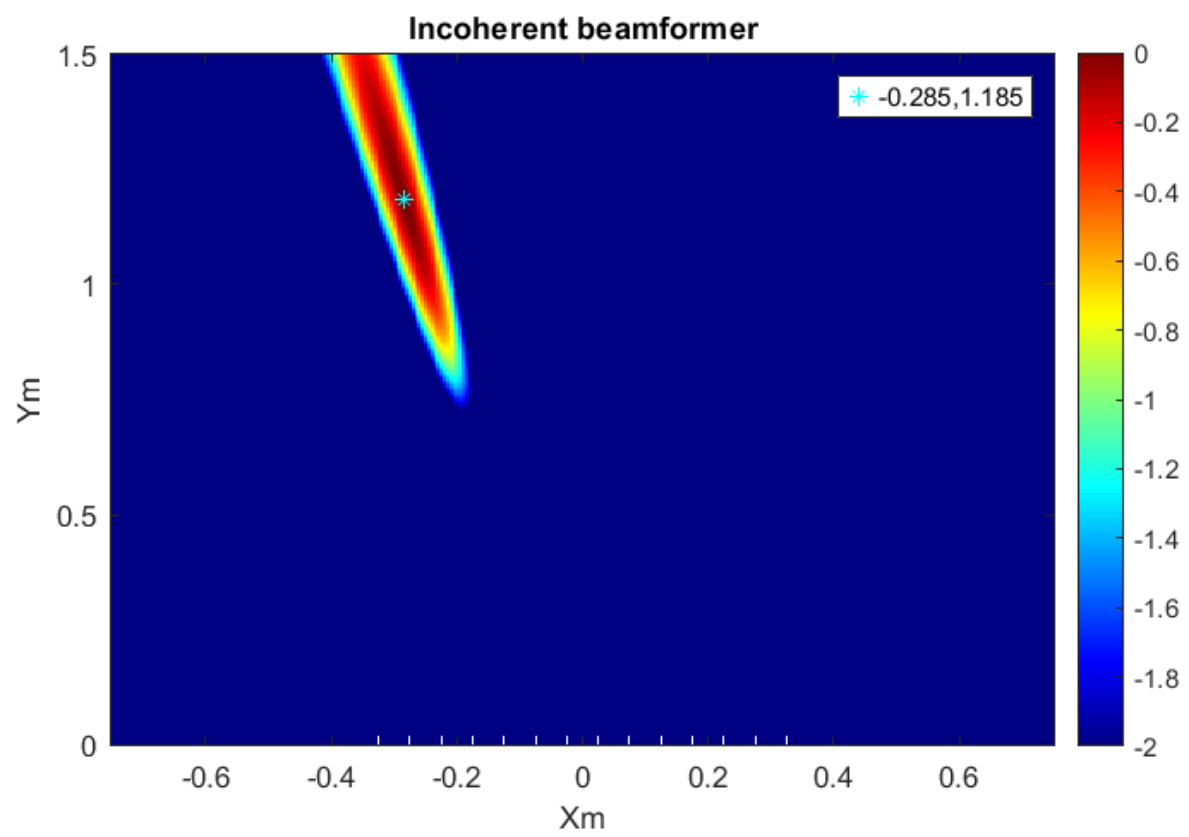
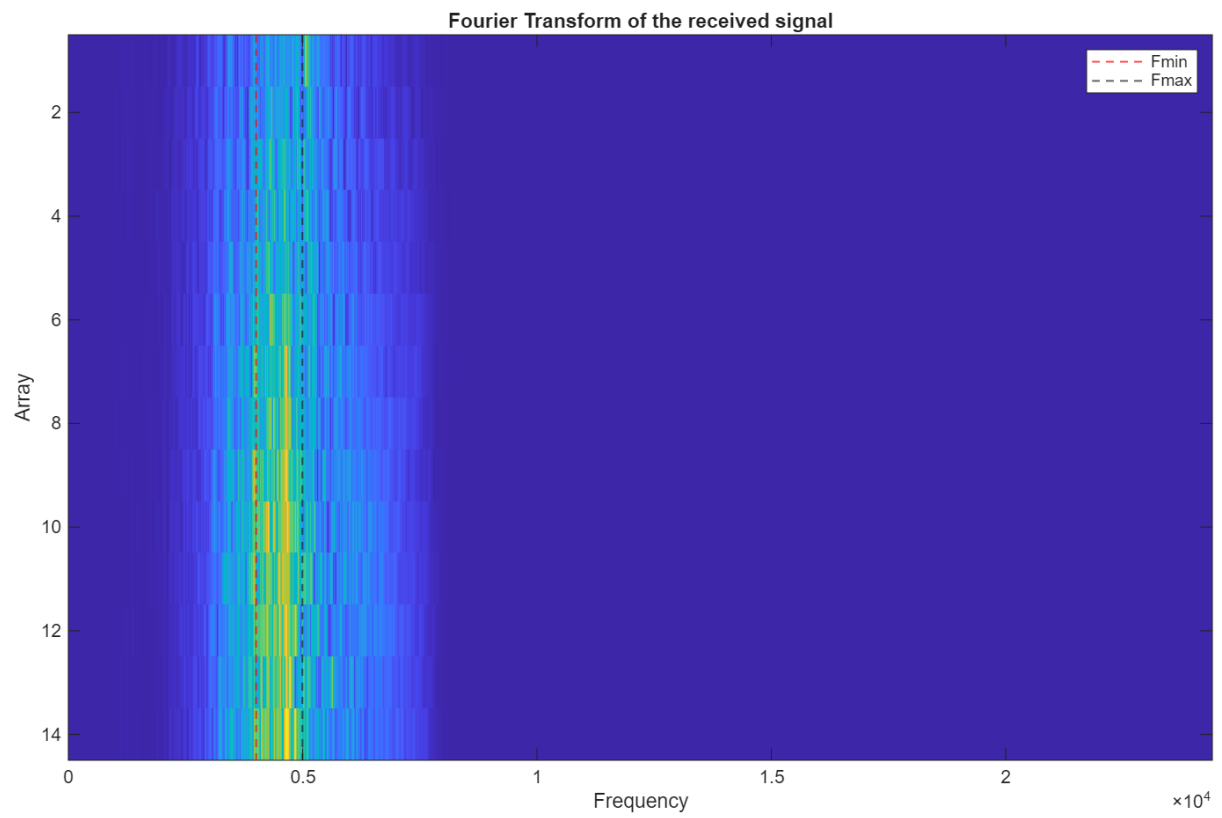
Using the given MATLAB codes and self-made codes, the data was processed to obtain the appropriate results.

Results

Figures for t2 and y2; longer time span and source at the distance more than 1m.

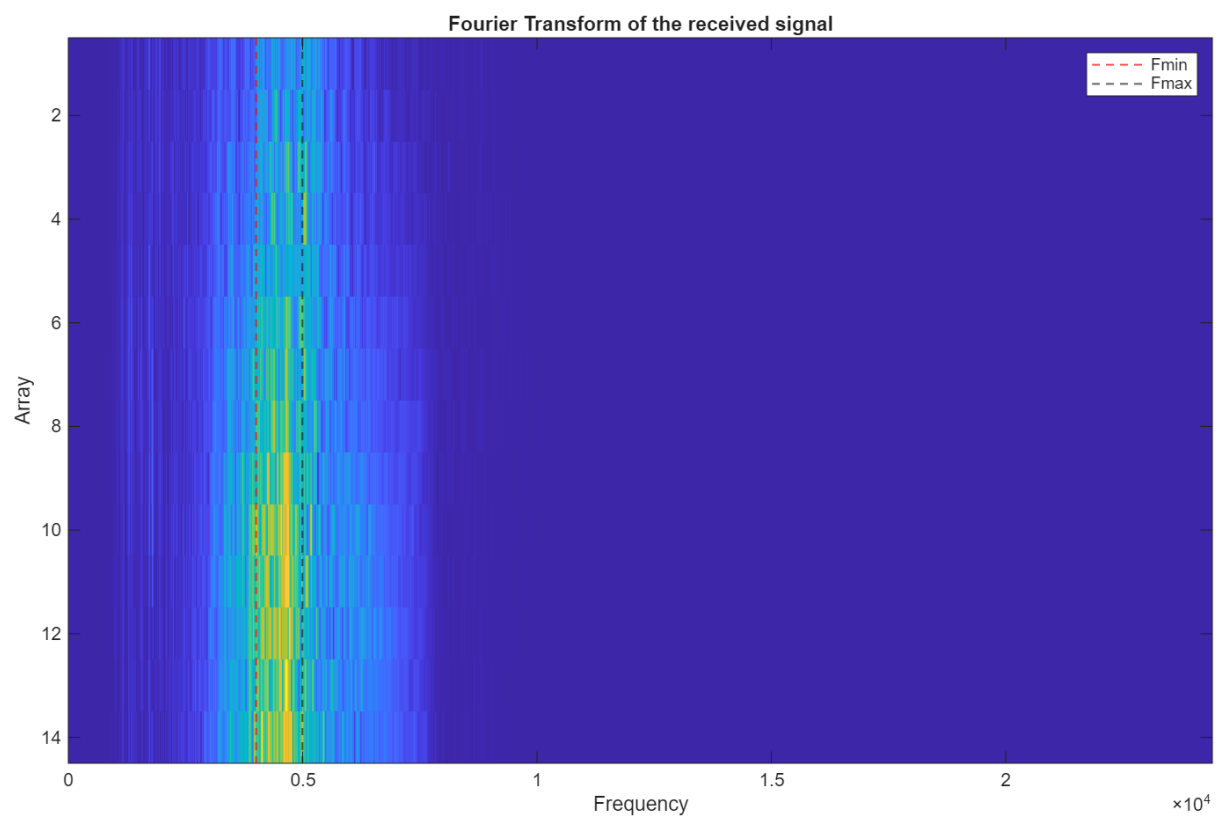
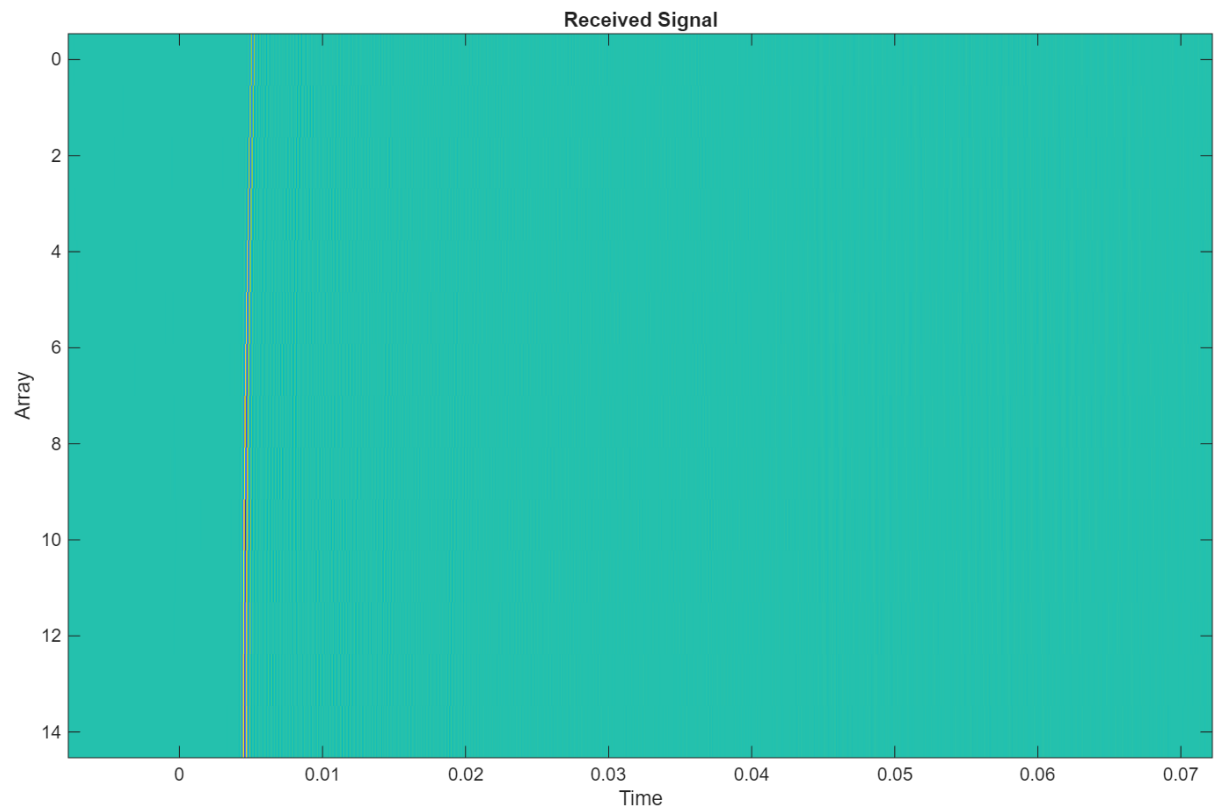
For y2 t2,

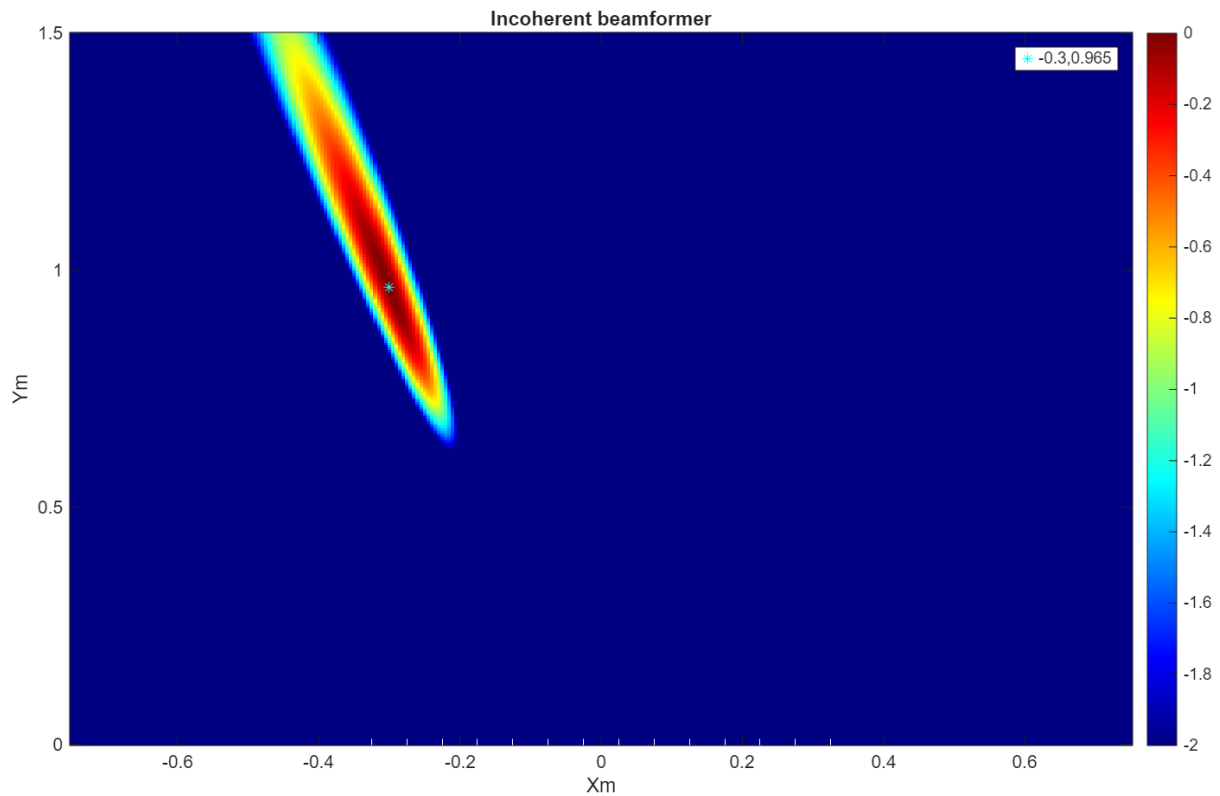




For y_1 t_1 ,

short time span and source at a distance less than 1m ($d = 90$ cm).





Interpretation of Result

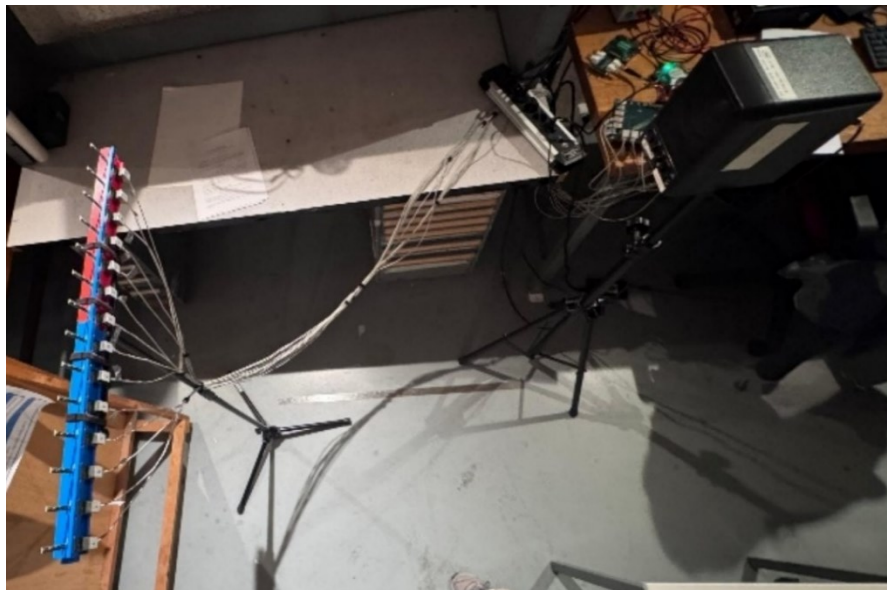
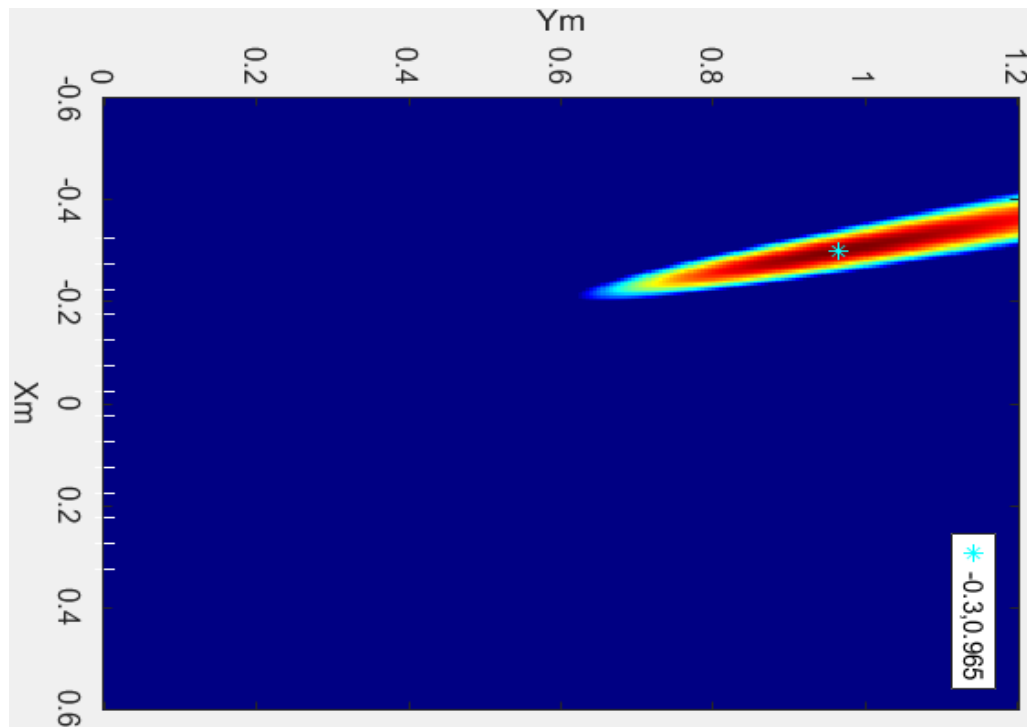
The experimental findings show the properties of the acoustic signals that the MEMS array receives and the system's capacity to use beamforming techniques to spatially locate the source. The study is separated into two parts: spatial focalization analysis and signal characterization in the time/frequency domains.

Plots of the received signal show the acoustic pressure measured over time across the array of 14 microphones. A distinct wavefront is observed arriving at around 5 ms. This delay corresponds to the time of flight for the sound to travel from the source to the array. The wavefront appears nearly vertically across the array indices, indicating that the source is located in the far field relative to the array aperture.

A Fast Fourier Transform (FFT) was used to confirm the incoming signals' spectral composition. From FFT, it is evident that the energy is focused on a small band that is centered between 4000 and 5000 Hz. The source characteristics, a Ricker/Gaussian signal with a core frequency of 4213 Hz, are exactly in line with this. This demonstrates that the particular frequency band of interest was successfully caught by the MEMS microphones without appreciable aliasing or distortion.

The energy of the back-propagated signals was projected across a 2D spatial grid (X_m , Y_m) to assess the system's spatial focusing capabilities, a key objective of Time Reversal

Acoustics. The geographic distribution of this concentrated energy is seen by the Incoherent Beamformer maps.



The focus areas are small in the X direction (cross-range/azimuth) and lengthy in the Y direction (range). Narrow in the x direction is because the aperture of the array extends along the X-axis, producing a constructive interference pattern that precisely specifies the angle of arrival. Similarly, in Y direction, there is poor depth resolution. This is because of the general feature of linear arrays in low-reverberation or free space settings. An Open TRM has a diffraction limit, as mentioned in the theory section. The axial focusing is less tight than the lateral focusing on the absence of a closed cavity or enough

multipath reflections to close the mirror. The elongation of the lobe also depends on the number of mics in the array as from a far enough distance the wave front almost becomes like that of a plane wave therefore increasing the number of mics and the total length of the array would improve the results giving better focusing.

In the case of **y1 and t1** we can confirm that the results are correct as the array is along the x-axis (represented by small white lines) and the middle of the array is the origin and the source is in the y direction. Therefore, by rotating the image and comparing it with the setup we can see that it matches.

The ruler on the ground is 1m and therefore the perpendicular distance of the source from the array is almost 1m. and the speaker is mounted a little ahead of the center of the stand where the distance measured is a little less than 1m.

It can also be seen that the speaker is in from of the microphone at the edge of the array which is -0.325m (as the total length is 0.65 and the origin is in the middle).

For y1, t1: The coordinates of the peak energy are $X = -0.3$ m, $Y = 0.965$ m. This matches the experimental setup designated " $y1 < 1$ m" and translates to a source distance of around one meter from the array.

For y2, t2: At coordinates $X = -0.285$ m, $Y = 1.19$ m, the energy is concentrated. This is for another source distance $y2 > 1$ m.

Conclusion

This lab experiment shows the effectiveness of Time Reversal Acoustics. At the source of origin, time reversal effectively compresses the scattered acoustic energy back into a brief pulse. Reverberation is advantageous in Time Reversal, in contrast to conventional imaging, where it is noise. Incorporating multipath reflections (Long Signal) enhances the focusing's spatial resolution and signal-to-noise ratio. The experiment demonstrates that a compact, finite array may get a focusing resolution equivalent to a considerably wider aperture in free space thanks to a complex medium.

For further improvements in the quality of focalization, experiments can have an increased number of microphones for improved spatial resolution. Also, the bandwidth of the source signal can be increased to have a sharp temporal peak.